Proposed Outline

Nanoparticle-Mediated Advances in Plant Tissue Culture: Global Research Evolution, Emerging Applications, and Future Directions—A Bibliometric Analysis

1. Introduction

1.1 Background

1.2 Nanoparticles in Plant Tissue Culture

1.3 Rationale for the Bibliometric Study

1.4 Research Gap

1.5 Aim

To quantitatively and visually analyze the global evolution, intellectual structure, collaboration patterns, emerging applications, and future research directions of nanoparticle-mediated plant tissue culture research using Scopus-indexed literature.

2. Research Objectives

Objective 1

To examine the **annual growth and global evolution** of research on nanoparticles in plant tissue culture.

Objective 2

To identify the **most productive and influential authors, institutions, countries and journals**.

Objective 3

To analyze **citation patterns and highly cited publications**.

Objective 4

To identify the major **nanoparticle types and plant tissue-culture applications** investigated.

Objective 5

To examine **author, institutional and country collaboration networks**.

Objective 6

To identify the **intellectual structure and thematic clusters** of the field.

Objective 7

To identify **emerging research themes and future research directions**.

3. Research Questions

1. How has nanoparticle-mediated plant tissue culture research evolved over time?
2. Which countries, institutions, authors and journals dominate the field?
3. Which publications have had the greatest scientific impact?
4. What are the major nanoparticle types investigated in plant tissue culture?
5. Which plant tissue-culture applications receive the greatest research attention?
6. What are the major collaboration patterns among countries, institutions and researchers?
7. What are the major intellectual and thematic clusters in this research domain?
8. Which topics are emerging and likely to shape future research?

4. Methodology

4.1 Study Design

- Bibliometric analysis
- Descriptive performance analysis
- Science mapping
- Network analysis
- Thematic analysis

4.2 Database Selection

Scopus will be used as the sole bibliographic database.

The methodology should explicitly state:

"Scopus will be selected as the sole bibliographic database because of its broad multidisciplinary coverage and extensive indexing of literature across plant sciences, biotechnology, nanotechnology, agricultural sciences, and materials science."

4.3 Search Strategy

The search should combine three major concepts:

Concept 1: Nanoparticles
Concept 2: Plant tissue culture
Concept 3: In-vitro culture/micropropagation

A preliminary Scopus strategy can be:

```
TITLE-ABS-KEY(
  nanoparticle* OR nanomaterial* OR "nano particle*" OR "nano-material*"
)
AND
TITLE-ABS-KEY(
  "plant tissue culture" OR "plant tissue cultures" OR
  "plant cell culture*" OR "plant cell suspension*" OR
  micropropagation OR micropropagation* OR
  "in vitro culture" OR "in vitro propagation" OR
  "in-vitro culture" OR "in-vitro propagation" OR
  organogenesis OR callogenesis OR
  "somatic embryogenesis" OR
  "hairy root culture*" OR
  "plant regeneration"
)
```

Important: Before finalizing the paper, the exact Scopus query will be tested and the final number of retrieved records reported because adding/removing terms will substantially change the dataset.

5. Eligibility Criteria

Inclusion Criteria

Exclusion Criteria

6. PRISMA-Based Study Selection

Even though this is a **bibliometric analysis rather than a conventional systematic review**, a PRISMA-style flow diagram can be used to transparently report the search and screening process.

Flow:

Scopus Search

↓

Records identified

↓

Duplicate records removed

↓

Records screened

↓

Records excluded

↓

Full-text/eligibility assessment

↓

Final bibliometric dataset

The final numbers should be taken directly from the Scopus export and screening process rather than predetermined.

7. Data Extraction

The following Scopus fields can be extracted:

- Author
- Title
- Year
- Journal
- Abstract
- Author keywords
- Indexed keywords
- Affiliations
- Countries
- Citations
- DOI
- Document type
- Subject area

- Funding information
- References
- Corresponding author
- International collaboration information

8. Bibliometric Performance Analysis

8.1 Annual Publication Growth

8.2 Citation Analysis

8.3 Most Productive Authors

8.4 Most Productive Journals

8.5 Leading Countries

8.6 Leading Institutions

9. Scientific Collaboration Analysis

9.1 Author Collaboration

9.2 Institutional Collaboration

9.3 Country Collaboration

10. Intellectual Structure of the Field

10.1 Co-Citation Analysis

10.2 Author Co-Citation

10.3 Bibliographic Coupling

11. Keyword and Thematic Analysis

This should be one of the **core sections** of your paper.

11.1 Keyword Co-Occurrence

Analyze:

- Author keywords
- Indexed keywords
- Keyword frequency
- Keyword clusters

11.2 Thematic Clusters

12. Emerging Research Applications

13. Thematic Evolution

14. Thematic Map

Using **Bibliometrix/Biblioshiny**, classify themes according to:

- **Motor themes**
- **Niche themes**
- **Basic themes**
- **Emerging/declining themes**

15. Trend Topics

Identify keywords showing increasing occurrence over time.

Potential emerging terms to investigate include:

- Green synthesis
- Nanomaterial
- Nanotoxicity
- Oxidative stress
- Secondary metabolites
- Genetic transformation
- CRISPR
- Gene delivery
- Smart delivery systems
- Sustainable micropropagation
- Plant regeneration

16. Future Research Directions

Based on the bibliometric findings, discuss:

16.1 Standardization

- Standardized nanoparticle characterization
- Consistent reporting of particle size and concentration
- Standardized experimental protocols

16.2 Nanotoxicity and Biosafety

- Dose-dependent toxicity
- Long-term effects
- Residual nanoparticles
- Environmental fate

16.3 Mechanistic Research

- ROS-mediated mechanisms
- Hormonal pathways
- Molecular signaling
- Nanoparticle–cell interactions

16.4 Precision Nanobiotechnology

- Targeted nanoparticle delivery
- Controlled release
- Smart nanomaterials

16.5 Sustainable Nanotechnology

- Green synthesis
- Biogenic nanoparticles
- Eco-friendly production

16.6 Industrial Scale-Up

- Large-scale micropropagation
- Cost-effectiveness
- Commercial production
- Reproducibility

17. Software and Visualization

Bibliometrix/Biblioshiny

For:

- Annual scientific production
- Sources

- Authors
- Countries
- Thematic evolution
- Thematic map
- Trend topics
- Three-field plot
- Conceptual structure

18. Results Structure

4.1 Search Results and Dataset Characteristics

4.2 Annual Scientific Production

4.3 Citation Analysis

4.4 Most Productive Authors

4.5 Most Influential Publications

4.6 Leading Journals

4.7 Leading Countries

4.8 Leading Institutions

4.9 Country Collaboration Network

4.10 Author Collaboration Network

4.11 Keyword Co-Occurrence Analysis

4.12 Co-Citation Analysis

4.13 Bibliographic Coupling

4.14 Thematic Mapping

4.15 Thematic Evolution

4.16 Emerging Research Trends

4.17 Future Research Frontiers

19. Discussion

5.1 Global Evolution of the Research Field

5.2 Dominant Research Contributors